AP CALCULUS AB

Unit 1 • Worksheet 1-2A

Worksheet 1-2A • Limits to a Point of a Function

Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

What is $\lim_{x \to 2} \left(3x^2 - x + 1\right)$?

(A) 11

(B) 13

(C) 9

(D) It does not exist.

Question 2

NO CALCULATOR

What is $\lim_{x \to -1} \left(x^3 + 2x\right)$?

(A) 3

(B) -3

(C) 1

(D) -1

Question 3

NO CALCULATOR

What is $\lim_{x \to 4} 7$?

(A) 4

(B) 0

(C) 7

(D) 28

Question 4

NO CALCULATOR

Let $p(x) = \begin{cases} x^2 - 3, & x < 2 \\ 2x - 3, & x \geq 2. \end{cases}$ What is $\lim_{x \to 2} p(x)$?

(A) It does not exist.

(B) 1

(C) 4

(D) -1

Question 5

NO CALCULATOR

Let $q(x) = \begin{cases} 5 - x, & x \leq 3 \\ x^2 - 6, & x > 3. \end{cases}$ What is $\lim_{x \to 3} q(x)$?

(A) 2
(B) 3
(C) It does not exist.
(D) $\frac{5}{2}$

Question 6

NO CALCULATOR

For the same q, what is $q(3)$?

(A) 2
(B) 3
(C) It does not exist.
(D) 5

Question 7

NO CALCULATOR

What is $\lim_{x \to 5} \frac{x^2 - 25}{x - 5}$?

(A) 0
(B) 10
(C) 5
(D) It does not exist.

Question 8

NO CALCULATOR

What is $\lim_{x \to -2} \frac{x^2 + 5x + 6}{x^2 - 4}$?

(A) $\frac{1}{4}$
(B) 0
(C) It does not exist.
(D) $-\frac{1}{4}$

Question 9

NO CALCULATOR

What is $\lim_{x \to 0} \frac{2x^2 + 6x}{x^2 - 3x}$?

(A) -2
(B) 2
(C) 0
(D) It does not exist.

Question 10 (Free response) NO CALCULATOR

(a) Find $\lim_{x \to 3} \dfrac{x^2 - x - 6}{x^2 - 9}$. Show the three steps.

(b) Let $h(x) = \begin{cases} x + 4, & x < 1 \\ x^2 + 4, & x \geq 1. \end{cases}$ Find $\lim_{x \to 1^-} h(x)$, $\lim_{x \to 1^+} h(x)$, $\lim_{x \to 1} h(x)$ and $h(1)$.

(c) A classmate says: "Plugging in gives $\frac{0}{0}$, so the limit is 0." Explain the mistake.